Any two h-initial natural-number algebras are equal, so being h-initial is a mere proposition

The Univalent Foundations Program

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Editorial note. The counted statement is the unlabelled author thm environment at source lines 434-437 and its proof environment opens on line 438, the next line; the pairing is by adjacency and the retained proof range 438-444 is the whole environment.

Editorial note. The environment carries no author label and the pinned book label-number dump has no entry for it, so no counter value is imposed and the counted theorem prints with the wrapper numbering; the original number is not claimed.

Editorial note. The counted statement and proof refer to no label outside the card: the retained ranges carry no cross reference command at all.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: any two h-initial natural-number algebras are equal (unlabelled author theorem opening at line 434).

Allowed standard prerequisites (editorial).

- Homotopy initial algebras. A natural-number algebra is homotopy initial when the type of algebra homomorphisms from it to any other such algebra is contractible.. It is the hypothesis of the statement and the source of the two homomorphisms used in the proof.
- Contractibility of a hom-type. In a contractible type any two elements are equal, and the type is inhabited.. It produces the two homomorphisms between the two algebras and makes their composites equal to the identities.
- Univalence and mere propositions. An equivalence of types gives an equality of them by univalence, and a type is a mere proposition when its identity types are contractible.. It turns the exhibited equivalence into the equality of the algebras and transfers the set-level claim to the property of being h-initial.

Theorem 0.0.1. *Any two h-initial* $\mathbb{N}$*-algebras are equal. Thus, the type of h-initial* $\mathbb{N}$*-algebras is a mere proposition.*

Proof. Suppose I and J are h-initial $\mathbb{N}$-algebras. Then $\mathbb{N}\mathsf{Hom}(I, J)$ is contractible, hence inhabited by some $\mathbb{N}$-homomorphism $f : I \to J$, and likewise we have an $\mathbb{N}$-homomorphism $g : J \to I$. Now the composite $g \circ f$ is a $\mathbb{N}$-homomorphism from I to I, as is id_I; but $\mathbb{N}\mathsf{Hom}(I, I)$ is contractible, so $g \circ f = \mathsf{id}_I$. Similarly, $f \circ g = \mathsf{id}_J$. Hence $I \simeq J$, and so $I = J$. Since being contractible is a mere proposition and dependent products preserve mere propositions, it follows that being h-initial is itself a mere proposition. Thus any two proofs that I (or J) is h-initial are necessarily equal, which finishes the proof. □